Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To eliminate the concern of Water leakage through door water-shield in Scorpio model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**10.05.19**	**10.05.19**
Completion Date	**20.11.19**	**27.12.19**

Current Status: **CLOSED**

Cross Functional Team

No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Nandy Tarit	L7-O	PRD	8975767608
2	**Mentor**	Chikhale Sandip	L5-M	STCF	9922408138
3	**Member 1**	Santosh Jadhav	L7-0	MQA	9423309400
4	**Member 2**	GANESH BENDALE	L7-O	PRD	9850952219
5	**Member 3**	Abhishek Sonawane	L8-0	STCF	9881751802
6	**Member 4**	Sachin Bhamare	L7-O	MQA	9158960111
7	**Member 5**	Kiran Badgujar	L7-0	SQA	9689922687

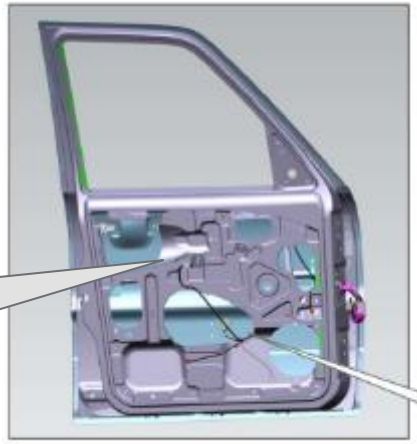
Step1. Select Subject

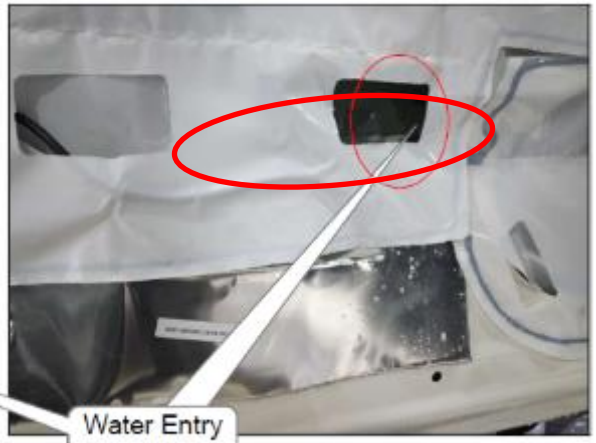
Brief description of Problem:-

To eliminate the concern of Water leakage through watershield in Scorpio model

What	:	Water leakage though door watershield.
Where	:	At Trim-3 Buyoff
When	:	Since Jan19
Who	:	By associate at Door trimming stage , Shower test .
WHY	:	High rework, FRC /FBO loss, Repeat issue at CAI-Shower test & GNOVAC Audit.
HOW	:	Avg. 86 cases/month reported at CAI-Shower Test and 2-3 cases at GNOVAC.

WATER ENTRY SOURCE _WATER-LEKAGE THROUGH DOOR WATER-SHIELD



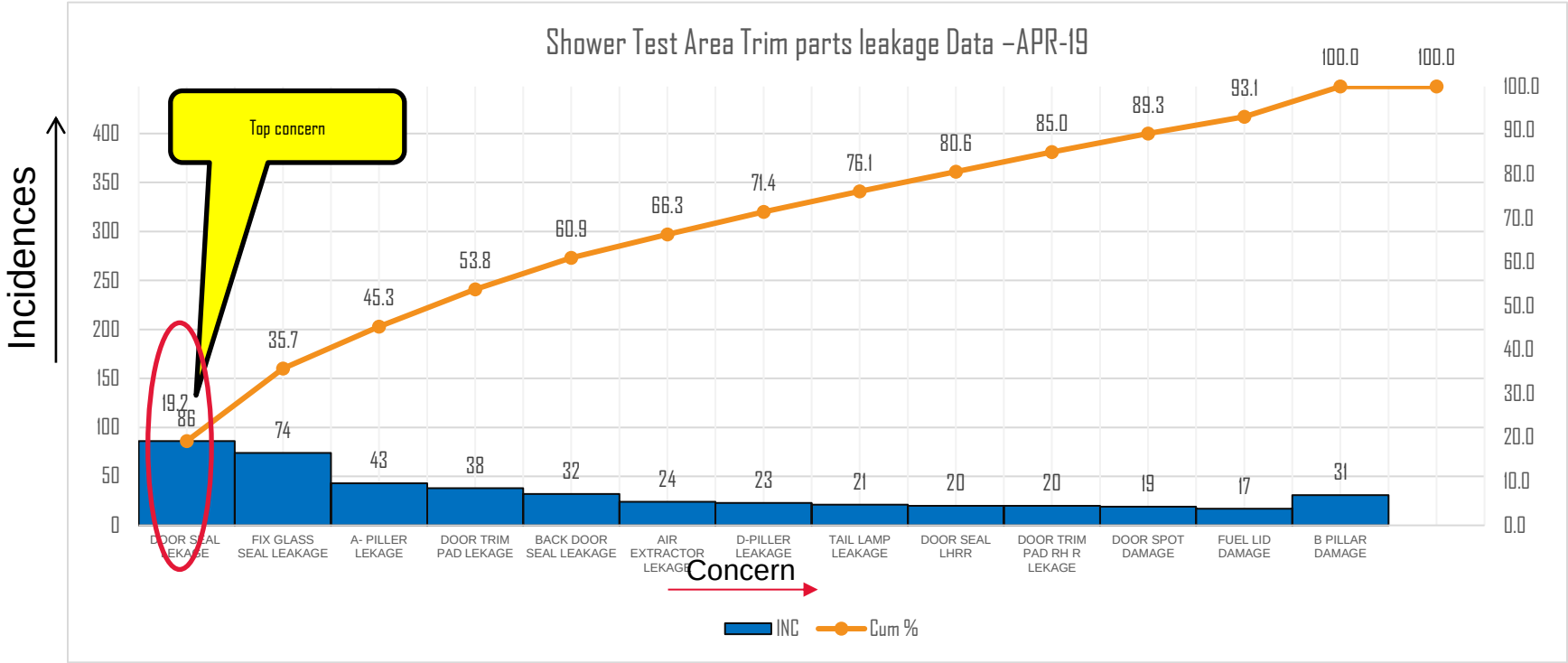


2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Inference:
Door seal Leakage due to watershield,is top concern and repeated chronic concern, hence taken for resolution

2. Justify the choice

Severity & Consequences of Concern :-

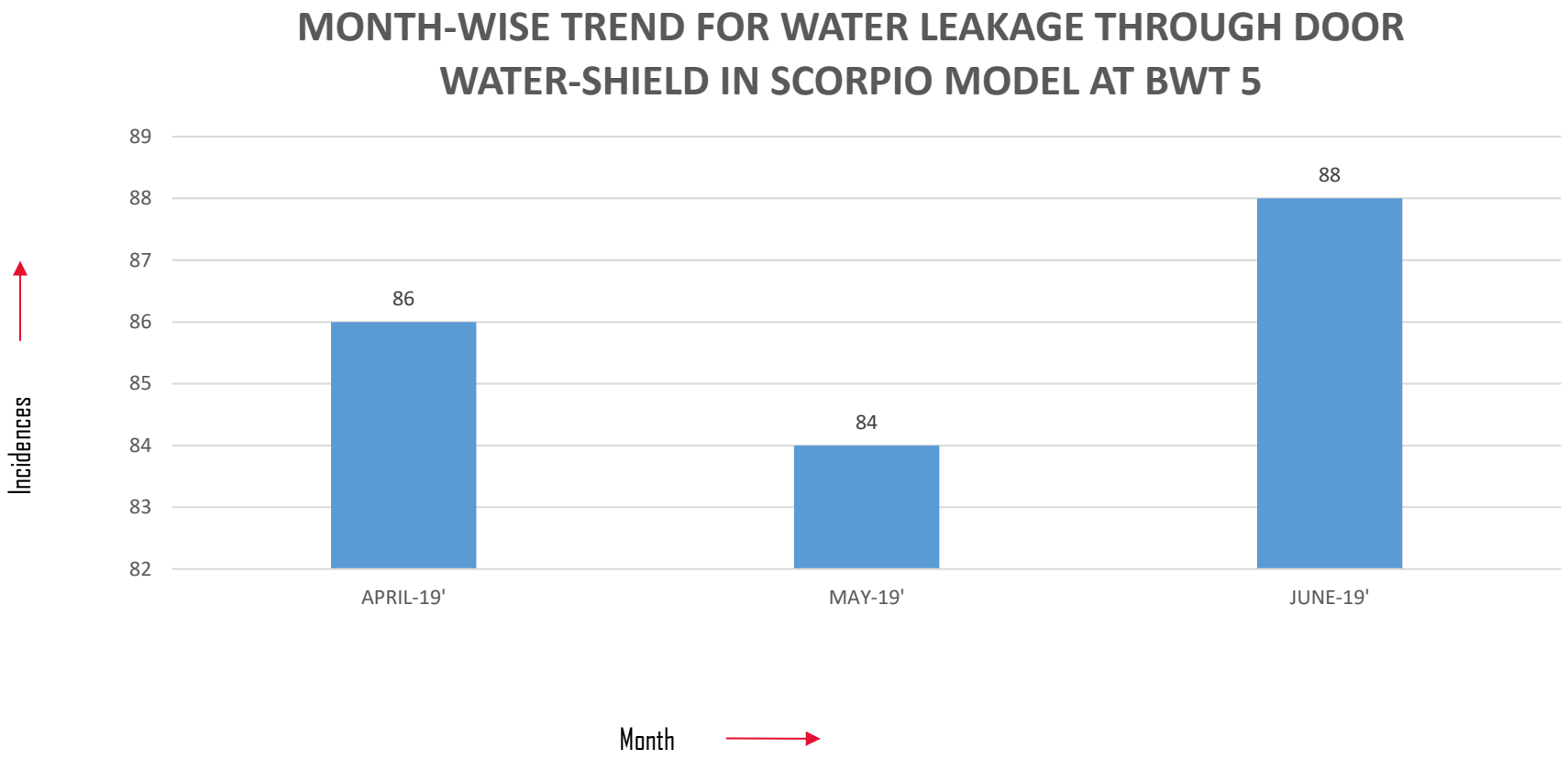
- **Avg. 84 Cases reported per month, for rework, 2-3 V1 concern reported in GNOVAC audit.**
- **Customer Dissatisfaction due to Water leakage.**
- **Incorrect appearance when customer enters or seat inside veh.**
- **Rework Difficult**
- **Rework cost more.**
- **Rework time 12 min (approx.) for each vehicle by 1 operator.**
- **Rework cost is 61130.3 Rs/year**

Cost of Poor Quality - Rework

Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	12
Avg Concerns/Day	Nos	4.0
Total Required Rework time per day	Min	48
Effective working time	Min	465
Operator required /day	nos	0.10
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	203.77
Avg no. of working days	days	25
Operator cost /month	Rs	5094.2
Operator Cost/year	**Rs**	**61130.3**

2. Justify the choice

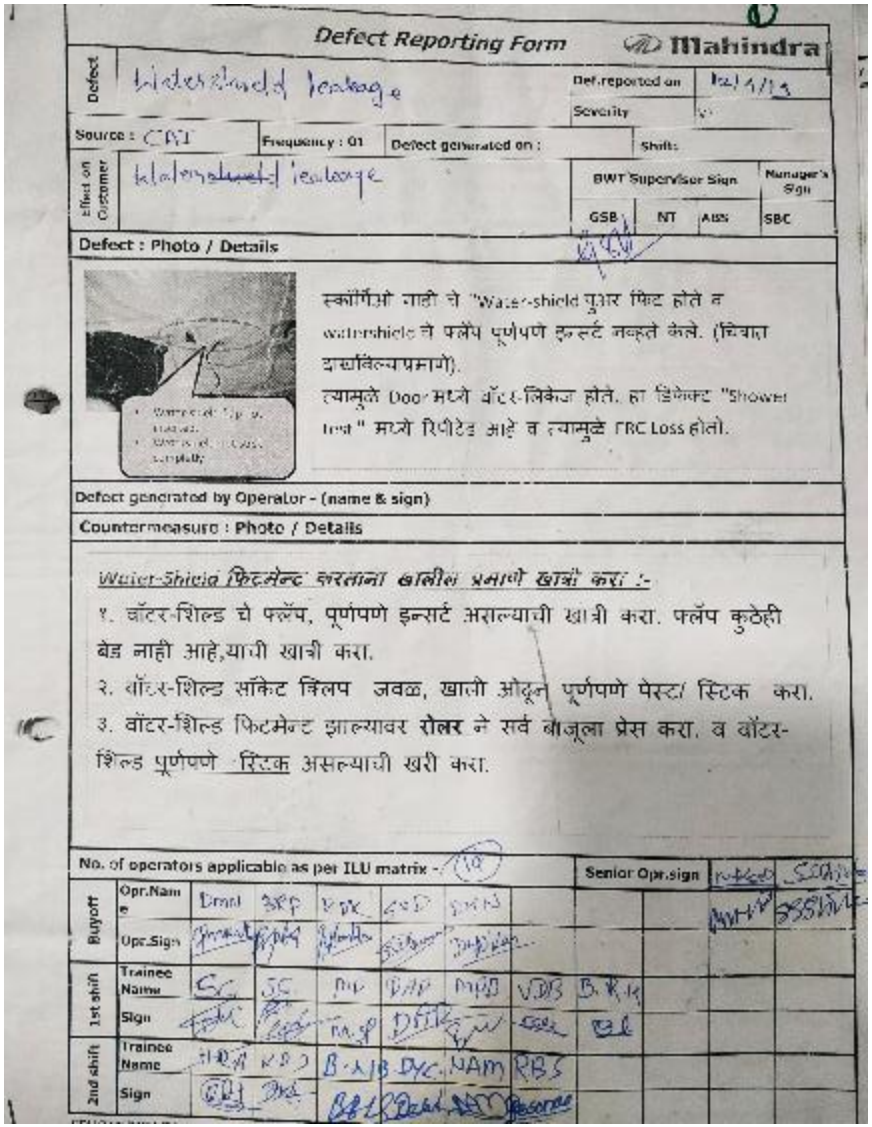
MONTHWISE DEFECT TREAD :-



Inference : Average 86 No.s concerns reported per month from April19 to June-19

2.b Immediate Containment Action

- Training for stage associates, explaining main step and key points.
- Emphasis on Watershield fitment as shown.
- 100% lot checked for supplier variability.
- DRF given to stage associates, explaining key points and focus on watershield fitment.



Defect Reporting Form — Mahindra

Defect: Watershield leakage

Def.reported on: 12/1/13

Severity:

Source: CPI | Frequency: 01 | Defect generated on: | Shift:

Effect on Customer: Watershield leakage

BWT Supervisor Sign | Manager's Sign

GSB | NT | ABS | SBC

Defect : Photo / Details

स्कॉर्पियो गाडी चे "Water-shield पूर्वार फिट होते व watershield चे फ्लॅप पूर्णपणे इन्सर्ट नव्हते केले. (चित्रात दाखविल्याप्रमाणे).
त्यामुळे Door मध्ये वॉटर-लिकेज होते. हा डिफेक्ट "Shower test" मध्ये रिपीटेड आहे व त्यामुळे FRC Loss होतो.

Defect generated by Operator - (name & sign)

Countermeasure : Photo / Details

Water-Shield फिटमेन्ट करताना खालील प्रमाणे खात्री करा :-

१. वॉटर-शिल्ड चे फ्लॅप, पूर्णपणे इन्सर्ट असल्याची खात्री करा. फ्लॅप कुठेही बेड नाही आहे,याची खात्री करा.

२. वॉटर-शिल्ड सॉकेट क्लिप जवळ, खाली ओढून पूर्णपणे पेस्ट/ स्टिक करा.

३. वॉटर-शिल्ड फिटमेन्ट झाल्यावर **रोलर** ने सर्व बाजूला प्रेस करा. व वॉटर-शिल्ड पूर्णपणे स्टिक असल्याची खरी करा.

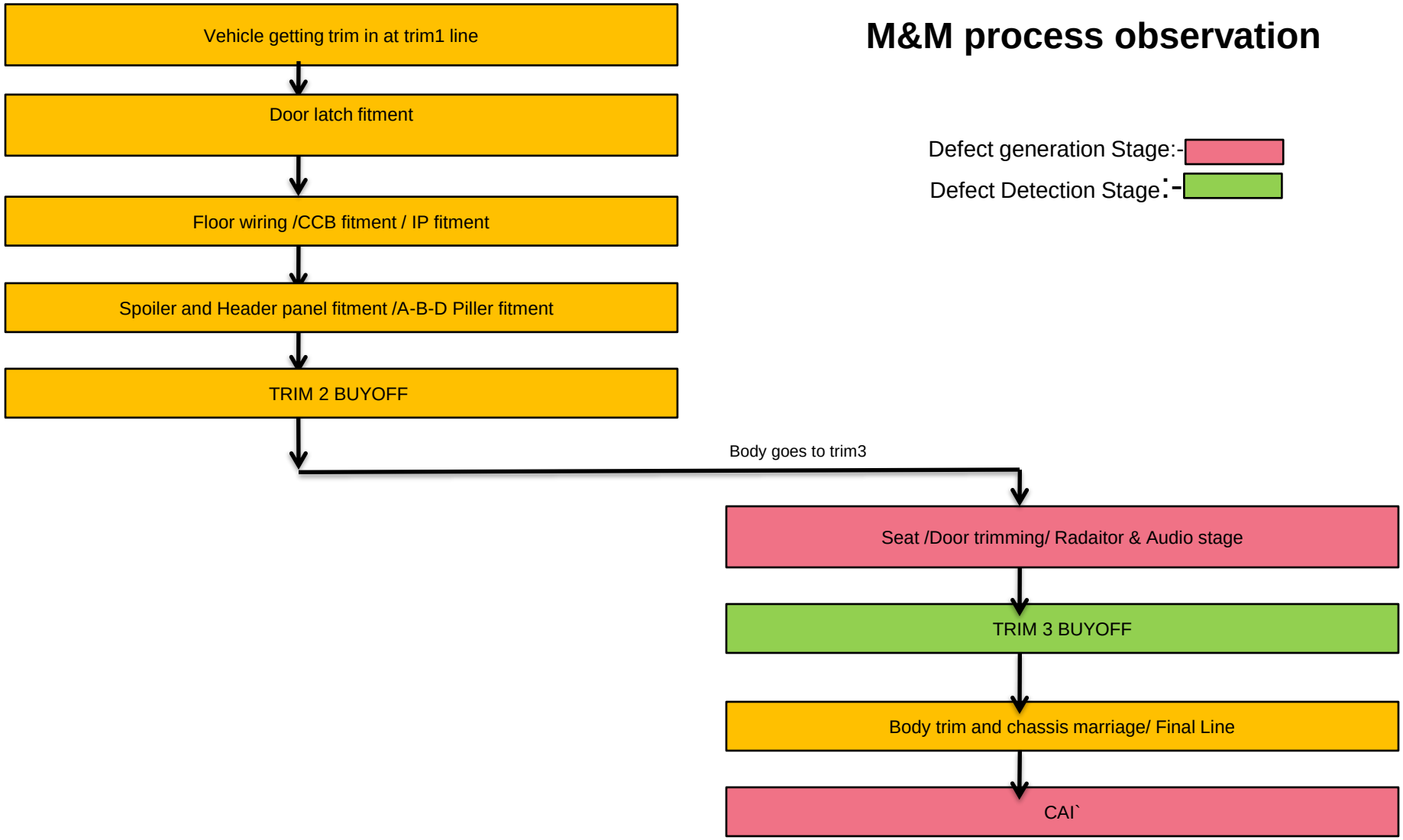
No. of operators applicable as per ILU matrix - 10

Senior Opr.sign

Buyoff	Opr.Name	Dmal	SRP	VDK	GSD	DMS					
	Opr.Sign										
1st shift	Trainee Name	G	JS	DV	DAP	DMD	VDB	B.K.R			
	Sign										
2nd shift	Trainee Name	HRA	KDJ	B.NB	DVC	NAM	RBS				
	Sign										

3. Understand the current situation

Flow Chart :- Observation of Process



Step 3. Understanding Current Situation

Observation of process

Vech at Water sheld stage

Silver Patch fitment

Speaker fitment

Water shield Fitment

Proper pasting at bottom

pressed by Roller

Door trim pad fitment

CHECKING AT SHOWER

Water-Shield Fitment at Door stage

Defect generation stage

Defect identification stage

Step 3. Understanding Current Situation

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area :SCORPIO/XYLO TCF
Cell / BWT / Line: BWT-5/TRIM3

Requested By- MQA | Date -25.4.19 | Source -GNOVAC | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : **Door seal leakage**

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- K2C36173 | Model : W110 S5

Date of Incidence: 22.3.19 | Assy : | Shift / Time : First shift | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☑ SC

Immediate containment action for Customer Protection :

DRF GIVEN TO ALL STAGE ASSOCAITE

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - CHIKHALE SANDEEP | BWT . : 6

Does a control already exist ? ☑ YES ☐ NO

Stage / Post Analysis -

Man(Human Action)

Operator Name : **Sonwane RB (APP)**
Is operator trained ? ☑ YES ☐ NO
Operator level (I.L.U) L
OR Dexterity level(1…4) L3
☐ YES ☑ NO
Is operator under training ? ☑ YES ☐ NO
Is operator temporary for Stage ? ☑ YES ☐ NO
Bad rework ? ☐ YES ☑ NO

Machine

Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☑ YES ☐ NO
Are machine parameters known ? (Record in Analysis table) ☐ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Method(process)

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP / SOS? ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Ope ? ☐ YES ☑ NO
Not applicable
In only one shift ? ☐ YES ☑ NO
In only one line ? ☑ YES ☐ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table) ☐ N/A ☐ NO

Product Failure

Other

Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
Is the packaging standard defined for the part ? ☑ YES ☐ NO
Is the packaging standard respected ? ☑ YES ☐ NO
Is there any alert mechanism for failure ? ☐ YES ☐ NO
N/A

Material/ Consumable

Is the specified product used ? ☐ YES ☐ NO
Did it evolve / Change ? ☐ YES ☐ NO
Is the storage SOP respected ? ☐ YES ☐ NO
Is the shelf life / expiry date respected ? ☐ YES ☐ NO
Is the part as per FIFO? ☐ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark

Following are the cuases and action plan for Door seal lekages :-

Sr.No	Causes	Action Plan	Owner	status
1	Poor pasting of water-shield	Daily Monitoring of Shower concerns and councelling with stage assocaite	Sandip Chikhale	On going
2	Water shield Flap not insert/Flap position not ok	ICA- Daily Monitoring of Shower concerns and councelling with stage assocaite. PCA-Not required Flap to be closed and tobe made as single water-shield	Sandip Chikhale	On going
3	Watershield NVH patch overlap	Implementation of Fixture for pasting NVH patch	Sandip Chikhale	CLOSED
4	Poor pasting of water-shield due to scoket clipping	ICA- Daily Monitoring of Shower concerns and councelling with stage associate. PCA-Water-shield position tobe shifted up-wards,above wiring clip,so drainage area will be incresed	Satish Pagar	OPEN
5	Watershild overlap with Flap	ICA- Daily Monitoring of Shower concerns and councelling with stage assocaite for correct flap pasting on door panel. PCA- Overlap flap to be Deleted	Satish Pagar	OPEN
6	Rear Door Fix glass seal	Taken up with Design	Satish Pagar	OPEN
7	Water entry into dor panel Gap between waistseal corner wr.t body	Fitment wise no issue observed. Source of water entry	Satish Pagar	Under monitoring
8	Water entry from ORVM triangular patch	Fitment wise no issue observed. Source of water entry	Satish Pagar	Under monitoring

Is 4M Analysis Part 2 required ? ☐ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☐ NO **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
Awareness given stage associate.(DRF GIVEN)	CHIKHALE SANDEEP	25.3.19	CLOSED

2. Permanent Action -	Resp	Date	Status
Action plan as above			

Analysed by Name/s - Chikhale Sandeep
Date - 25.4.19

Choice : ☐ WSS ☐ QC Story

L0 ANALYSIS FOR MIRROR PATCH CAP POOR FIT (S2/S4)

Inference from L0
Problem follows part & vehicle
Problem is a repeat problem
L1 analysis required

☑ V1

4M analysis Good and Not so good observations	
MAN	Operator dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows part and vehicle
Material	Handling and shelf life of part adhered

NO QUICK WIN

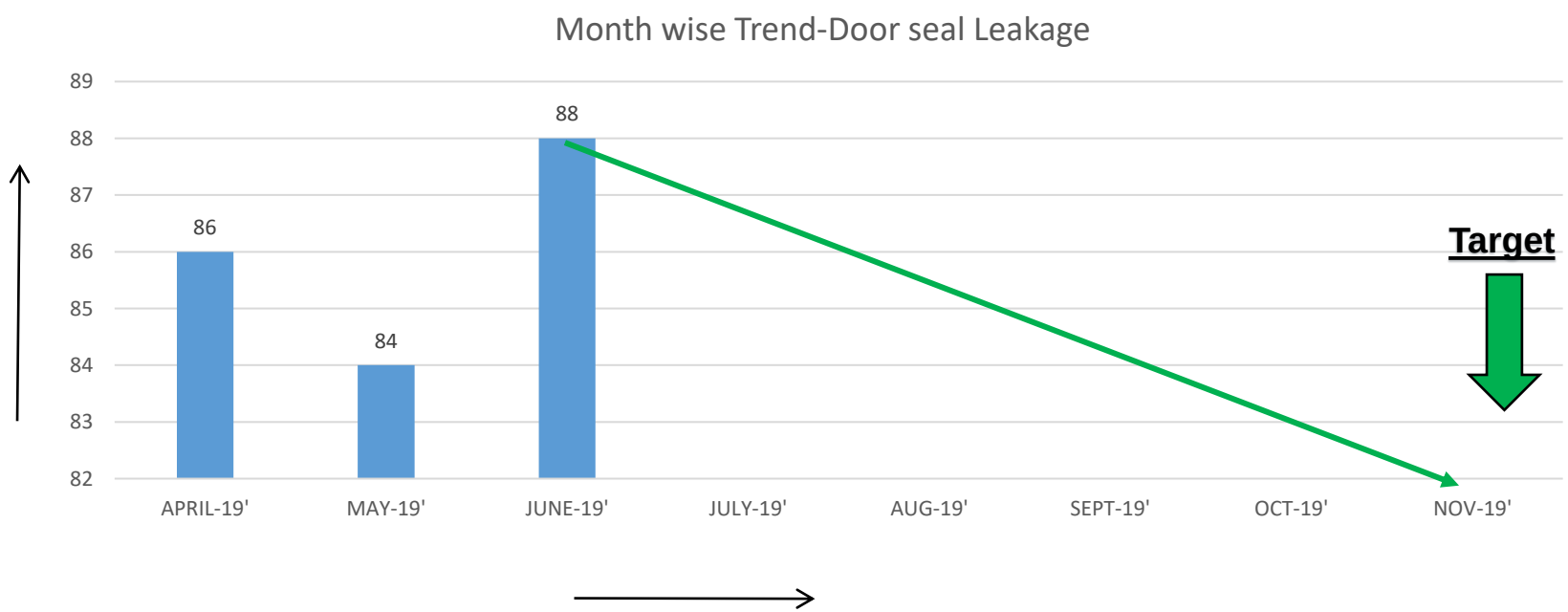
Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed only in Scorpio/Marazzo Model.
- Issue reported in both the shifts
- Repair time is high, around 8 minutes per vehicle
- Level 3 operators are deployed for Door trimming fitment.
- Operators are aware of key point & check point at respective stages.
- WaterShield is getting poor fit in Scorpio vehicles.

4. Select targets

Present Level:	**Avg. 84 Incidences /month are being reported at CAI**
Target Level:	**Elimination of concern Water-leakage thorugh water shield by end of Nov -19.**



Step 5. Analysis

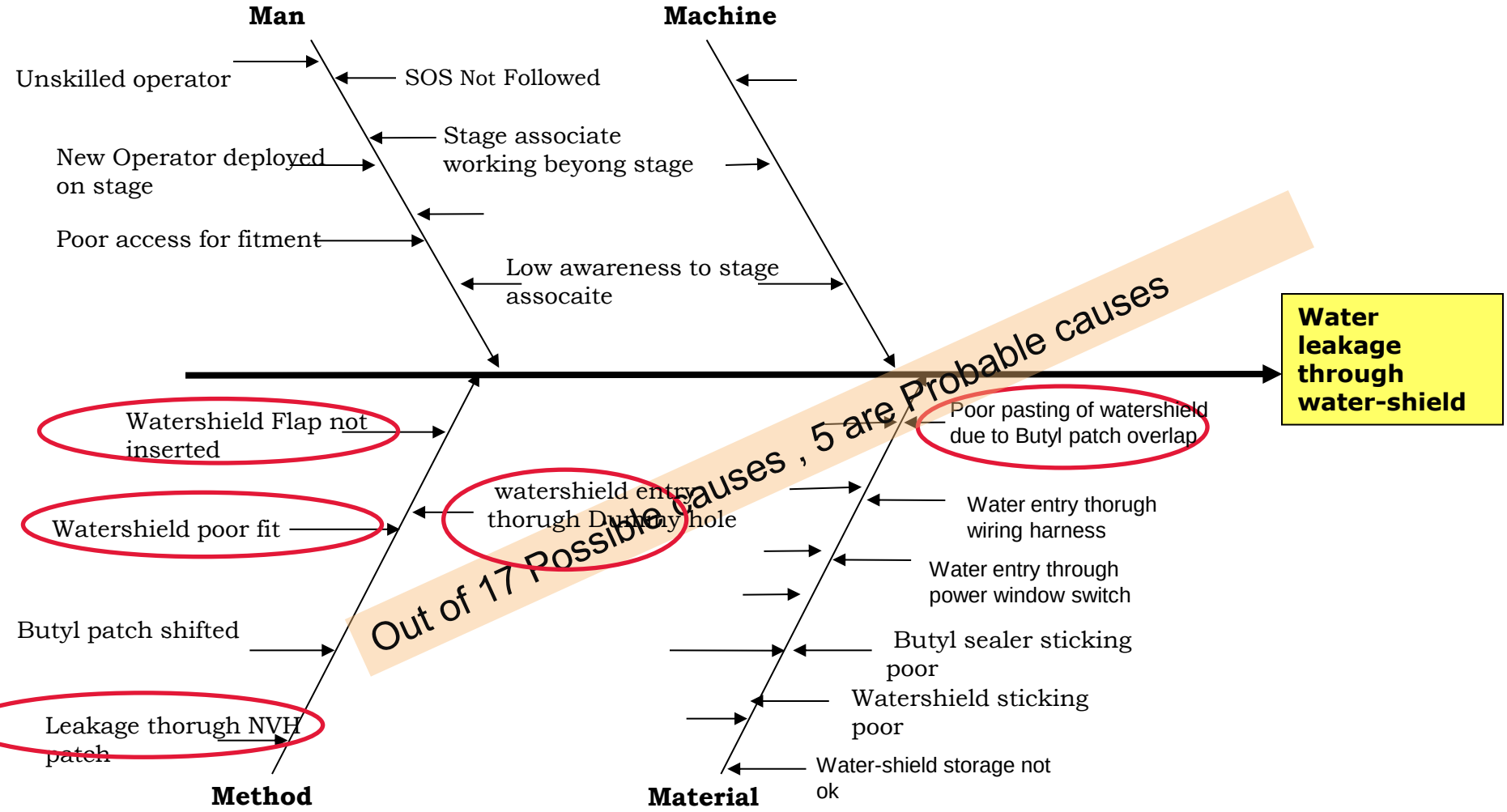
List of Possible causes identified in Brain storming session

Brainstorming Done along with CFT & we have found out 18 Possible Causes for this defect.

S N	Probable Cause
1	Unskilled associate
2	New associate deployed on stage
3	SOS Not Followed
4	Low awareness to stage assocaite
5	Stage assocaite working beyond stage
6	Watershield flap not inserted
7	Poor access for fitment
8	Leakage through NVH patch
9	water entry thorugh wiring harness
10	water entry thorugh power window switch
11	Butyl patch shifted
12	water entry thorugh window regulator clip
13	watershield poor fit
14	Water shield entry thorugh dummy hole (Model: SCORPIO S3)
15	Poor pasting of water shield due to Butyl patch overlap.
16	watershield sticking poor
17	watershield butyl sealer shifted
18	watershield storage not ok

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Probable causes :

S N	Probable Cause	Ganesh	Abhishek	Santosh	Satish	Nandy	SS Shelke	Total
1	Unskilled associate	**3**	**1**	**3**	**0**	**1**	**1**	**9**
2	New associate deployed on stage	**3**	**3**	**3**	**3**	**3**	**1**	**16**
3	SOS Not Followed	**0**	**1**	**0**	**0**	**1**	**3**	**5**
4	Low awareness to stage assocaite	**1**	**0**	**3**	**1**	**3**	**0**	**8**
5	Stage assocaite working beyond stage	**0**	**0**	**0**	**1**	**0**	**1**	**2**
6	Watershield flap not inserted	**9**	**3**	**9**	**3**	**3**	**9**	36
7	Poor access for fitment	**0**	**0**	**0**	**1**	**0**	**1**	**2**
8	Leakage through NVH patch	**9**	**3**	**3**	**9**	**3**	**9**	**36**
9	water entry thorugh wiring harness	**1**	**3**	**0**	**1**	**1**	**0**	**6**
10	water entry thorugh power window switch	**0**	**3**	**1**	**3**	**0**	**3**	**10**
11	Butyl patch shifted	**3**	**3**	**0**	**3**	**3**	**3**	**15**
12	water entry thorugh window regulator clip	**3**	**3**	**3**	**1**	**1**	**3**	**14**
13	watershield poor fit	**9**	**3**	**3**	**9**	**3**	**9**	**36**
14	Water shield entry thorugh dummy hole (Model: SCORPIO S3)	**9**	**9**	**9**	**3**	**9**	**3**	**42**
15	Poor pasting of water shield due to Butyl patch overlap.	**9**	**9**	**9**	**9**	**9**	**9**	**54**
16	watershield sticking poor	**0**	**1**	**0**	**3**	**1**	**3**	**8**
17	watershield butyl sealer shifted	**3**	**1**	**0**	**3**	**1**	**3**	**11**
18	watershield storage not ok	**3**	**1**	**0**	**3**	**1**	**3**	**11**

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 36.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering.

S N	Probable Cause	Reason
1	Unskilled associate	**All Trained assocaites**
2	New associate deployed on stage	**All Trained assocaites**
3	SOS Not Followed	**SOS are followed**
4	Low awareness to stage assocaite	**Found ok**
5	Stage assocaite working beyond stage	**Awareness given to stage assocaite**
6	Poor access for fitment	**20 veh checked found ok**
7	water entry thorugh wiring harness	**30 veh checked found ok**
8	water entry thorugh power window switch	**Lineside stock checked-found ok**
9	Butyl patch shifted	**Visually checked_Found ok**
10	water entry thorugh window regulator clip	**Visually checked_Found ok**
11	watershield sticking poor	**Butyl sealer checked-found ok**
12	watershield butyl sealer shifted	**Visually checked_Found ok**
13	watershield storage not ok	**Lineside stock checked-found ok**

5.1 Identify Probable cause

S N	Probable Cause
1	Watershield flap not inserted
2	Leakage through NVH patch
3	watershield poor fit
4	Water shield entry thorugh dummy hole (Model: SCORPIO S3)
5	Poor pasting of water shield due to Butyl patch overlap.

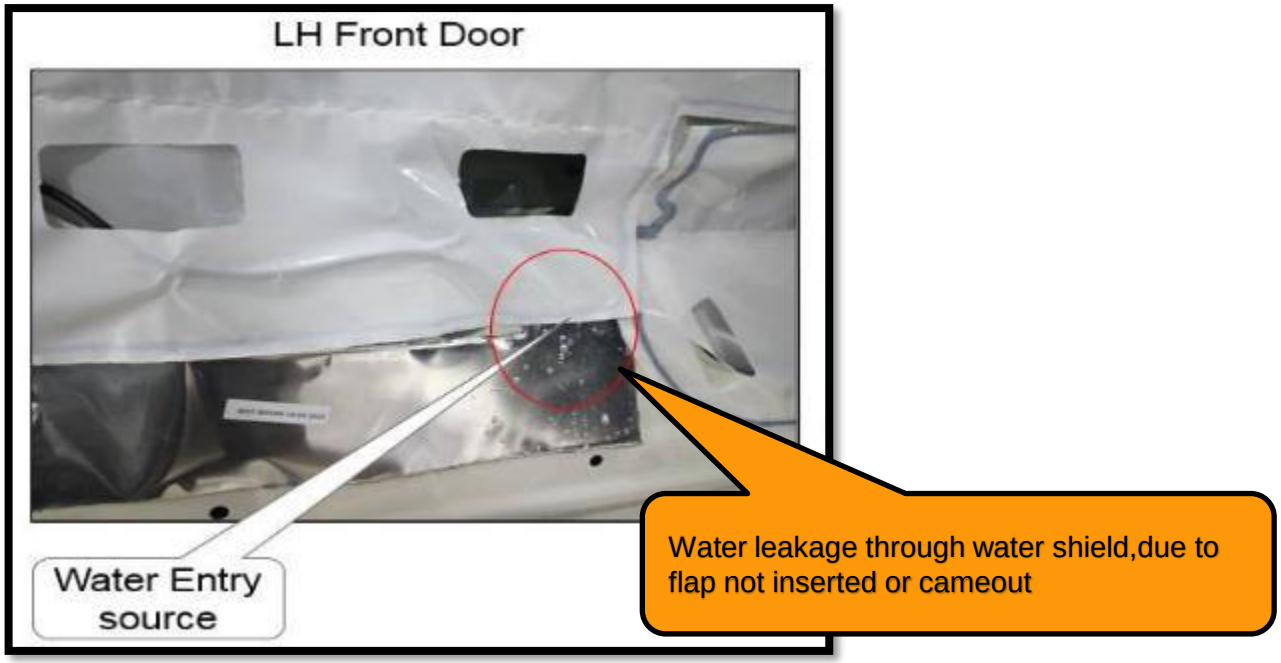
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Water-Shield Flap not inserted	Visual check	TOTAL =25 Vehicle checked, it was found that if water-shield flap is not inserted completely, water ingress into door trim. This result in to water lekage	Hypothesis valid

WATER-SHIELD FLAP NOT INSERTED

Contributors :-



WATER-SHILED FALP INSERTION PROCESS

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Water-Shield Flap not inserted	Visual check	TOTAL =25 Vehicle checked, it was found that if water-shield flap is not inserted completely, water ingress into door trim. This result in to water lekage	Hypothesis valid

WATERSHIELD FITMENT (FLAP INSERTION DATA)

Sr No	Vin No	Model	REMARKS
1	MA1TA2TDKK2E49492	W110 - S3 2WD LCCR - ABS - BS4 - 9 str	OK
2	MA1TA2WR2K2E49504	W110 S11 2WD 7SF ABS A/BAG 140HP	OK
3	MA1TA2TDKK2E49493	W110 - S3 2WD LCCR - ABS - BS4 - 9 str	NOT OK
4	MA1TA2WR2K2E49505	W110 S11 2WD 7SF ABS A/BAG 140HP	OK
5	MA1TA2TDKK2E49494	W110 - S3 2WD LCCR - ABS - BS4 - 9 str	OK
6	MA1TA2TDKK2E49486	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	NOT OK
7	MA1TA2TDKK2E49495	W110 - S3 2WD LCCR - ABS - BS4 - 9 str	OK
8	MA1TA2TDKK2E49473	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
9	MA1TA2WR2K2E49507	W110 S11 2WD 7SF ABS A/BAG 140HP	OK
10	MA1TA2SJXK2E49496	W110 S5 2WD 7SF ABS A/BAG 120HP	OK
11	MA1TA2TDKK2E49470	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	NOT OK
12	MA1TA2TDKK2E49469	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
13	MA1TA2TDKK2E49474	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
14	MA1TA2TDKK2E49471	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	NOT OK
15	MA1TA2WR2K2E49452	W110 S11 2WD 7SF ABS A/BAG 140HP	NOT OK
16	MA1TA2TDKK2E49475	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
17	MA1TA2SJXK2E49497	W110 S5 2WD 7SF ABS A/BAG 120HP	OK
18	MA1TA2TDKK2E49488	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
19	MA1TA2WR2K2E49453	W110 S11 2WD 7SF ABS A/BAG 140HP	NOT OK
20	MA1TA2WR2K2E49509	W110 S11 2WD 7CC ABS A/BAG 140HP	OK
21	MA1TA2TDKK2E49476	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
22	MA1TA2WR2K2E49454	W110 S11 2WD 7SF ABS A/BAG 140HP	OK
23	MA1TA2TDKK2E49489	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
24	MA1TA2TDKK2E49473	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK
25	MA1TA2TDKK2E49474	W110 - S3 2WD LCCR - ABS - BS4 - 7SF	OK

AS PER DATA,TOTAL 25 VEHICLE CHECKED, 6 VEHICLES FOUND NOT OK DUE TO FLAP INCORRECT POSITION

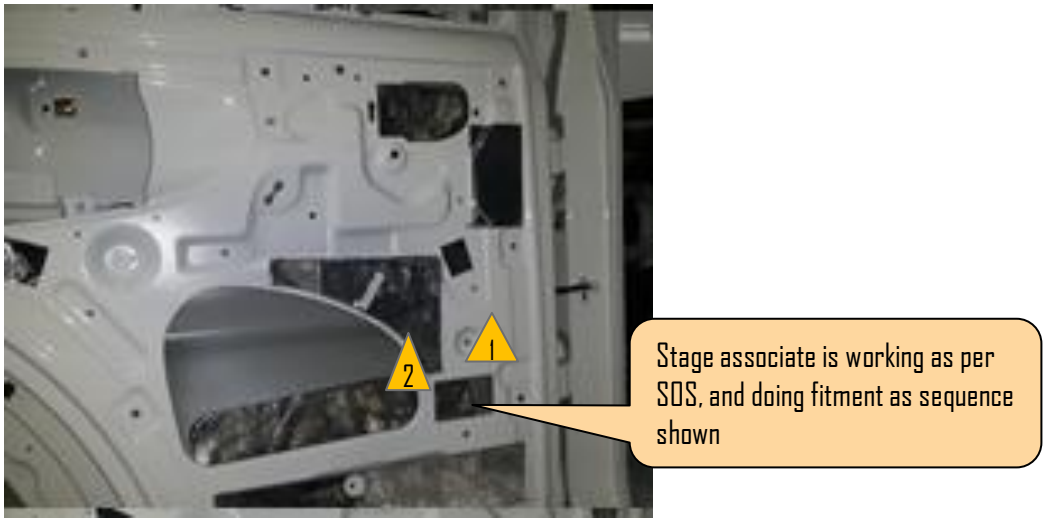
Conclusion:

It is observed that as watershield flap is not inserted completed or position id incorrect (bend).This is resulting in to water entery ad hence lekage

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Leakage thorugh NVH patch	Visual check	Total=25veh checked, Process checked for NVH Patch. It was observed that,leakage from NVH patch topside.	**Hypothesis Valid**



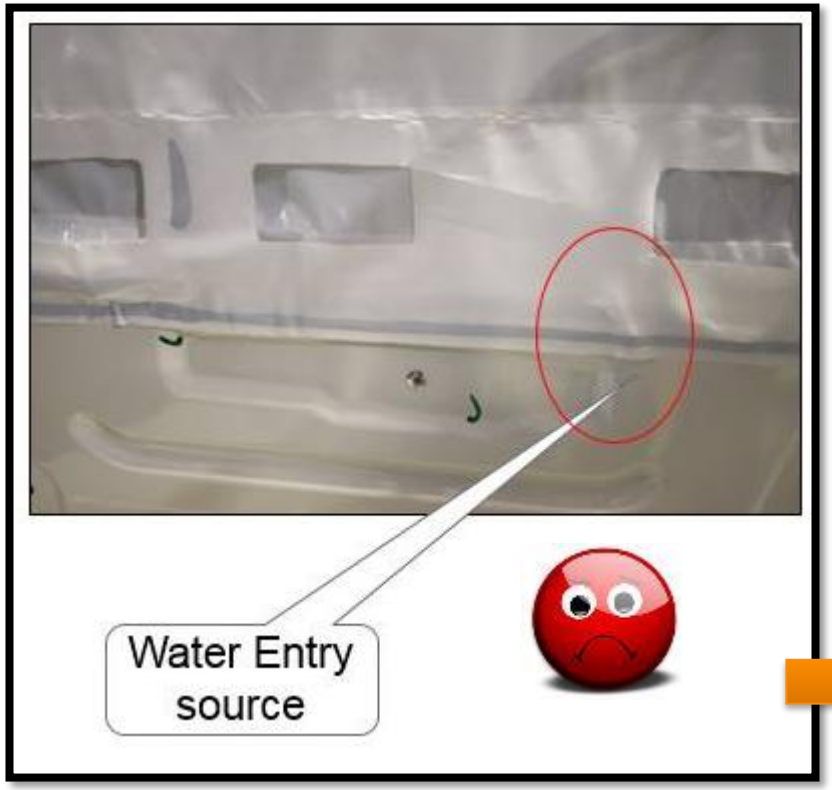
SR. NO	VIN No.	Model Description	RH FT
1	MA1TA2SJXK2H23284	W110 S5 2WD 9SF ABS A/BAG 120HP	✓
2	MA1TA2TDKK2H23297	W110 - S3 2WD LCCR - ABS - AB - BS4 - 9 str	✓
3	MA1TA2WR2K2H23298	W110 S9 2WD 7SF ABS A/BAG 140HP	✓
4	MA1TA2TDKK2H23277	W110 - S3 2WD LCCR - ABS - AB - BS4 - 7SF	✓
5	MA1TA2WR2K2H23243	W110 S11 2WD 7SF ABS A/BAG 140HP	✓
6	MA1TA2TDKK2H23262	W110 - S3 2WD LCCR - ABS - AB - BS4 - 9 str	x
7	MA1TA2WR2K2H23299	W110 S9 2WD 7SF ABS A/BAG 140HP	x
8	MA1TA2TDKK2H23260	W110 - S3 2WD LCCR - ABS - AB - BS4 - 7SF	✓
9	MA1TA2WR2K2H23244	W110 S11 2WD 7CC ABS A/BAG 140HP	x
10	MA1TA2TDKK2H23262	W110 - S3 2WD LCCR - ABS - AB - BS4 - 9 str	x
11	MA1TA2WR2K2H23300	W110 S9 2WD 7SF ABS A/BAG 140HP	x
12	MA1TA2SJXK2H23287	W110 S5 2WD 9SF ABS A/BAG 120HP	✓
13	MA1TA2TDKK2H23279	W110 - S3 2WD LCCR - ABS - AB - BS4 - 9 str	✓
14	MA1TA2WR2K2H23245	W110 S11 2WD 7SF ABS A/BAG 140HP	x
15	MA1TA2WR2K2H23301	W110 S9 2WD 7SF ABS A/BAG 140HP	x
16	MA1TA2WR2K2H23302	W110 S9 2WD 7SF ABS A/BAG 140HP	✓
17	MA1TA2SJXK2H23289	W110 S5 2WD 9SF ABS A/BAG 120HP	✓
18	MA1TA2SJXK2H23280	W110 S5 2WD 7SF ABS A/BAG 120HP	x
19	MA1TA2SJXK2H23246	W110 S5 2WD 9SF ABS A/BAG 120HP	x
20	MA1TA2TDKK2H23265	W110 - S3 2WD LCCR - ABS - AB - BS4 - 7SF	✓
21	MA1TA2TDKK2H23281	W110 - S3 2WD LCCR - ABS - AB - BS4 - 9 str	x
22	MA1TA2TDKK2H23265	W110 - S3 2WD LCCR - ABS - AB - BS4 - 7SF	x
23	MA1TA2SJXK2H23290	W110 S5 2WD 9SF ABS A/BAG 120HP	x
24	MA1TA2TDKK2H23266	W110 - S3 2WD LCCR - ABS - AB - BS4 - 7SF	✓
25	MA1TA2TDKK2H23266	W110 - S3 2WD LCCR - ABS - AB - BS4 - 7SF	x
26	MA1TA2TDKK2H23248	W110 - S3 2WD LCCR - ABS - AB - BS4 - 9 str	✓
27	MA1TA2SJXK2H23292	W110 S5 2WD 7SF ABS A/BAG 120HP	✓
28	MA1TA2SJXK2H23268	W110 S5 2WD 9SF ABS A/BAG 120HP	x

Conclusion –
Stage associate followed the SOS correctly & ensures correct fitment process.
Till 4 Vehicles reported at shower test for water lekage through NVH patch

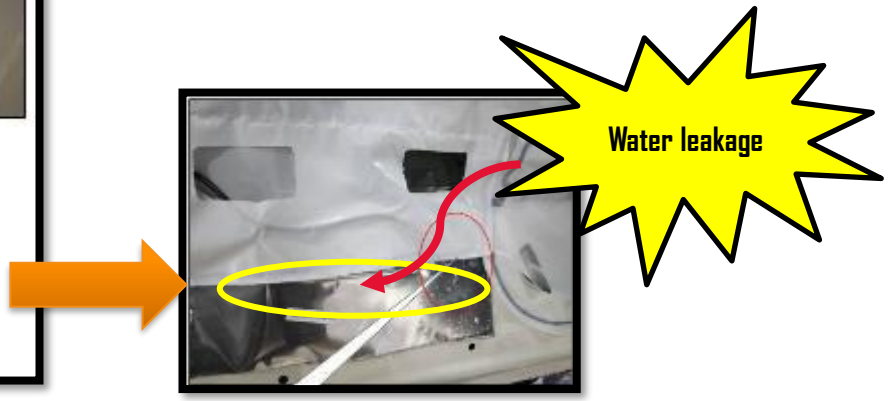
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Watershield poor fit	Visual check	At Gemba Door stage,it was observed,stage associate not pressing watershield completely to door panel . This results into water leakage	Hypothesis Valid



Conclusion :
At Gemba, due to water-shield butyl sealer is not pressed completely to door panel ,this results into water leakage.Hypothesis is Valid.

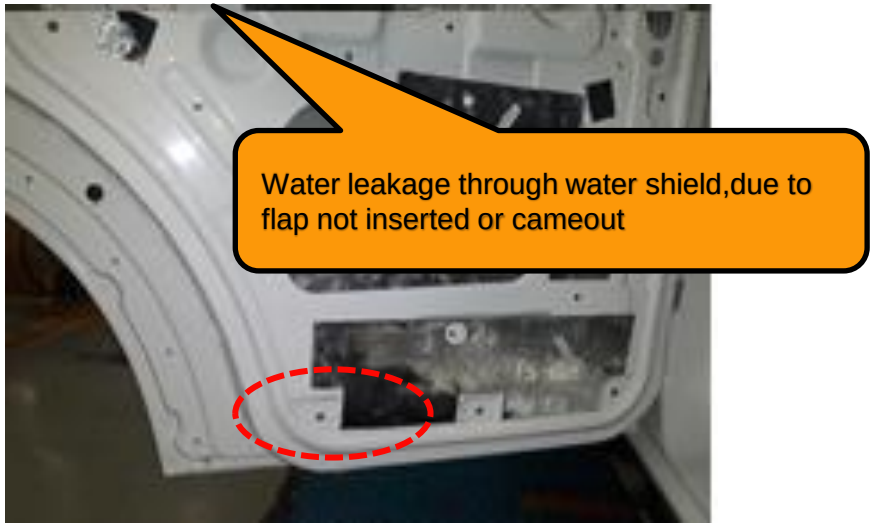


TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Water shield entry thorugh dummy hole (Model: SCORPIO S3)	Visual check	It was found that, Model Specific –Scorpio-S3,dummy hole for locator insertion of high variant door trim. This result into waterlekage inside vehicle	Hypothesis valid

COMPARISON STUDY OF DOOR TRIM PAD :-



WATER ENTRY THROUGH DOOR TRIM MTG DUMMY HOLE (MODEL-S3)

ANALYSIS:

1.For Lower variants Door trim, there is no locator for door trim.

2.For High Variant Molded door trim.there is locator at bottom door trim.This locator is getting inserted into mounting hole as shown.

3. As there is no locator for Lower varints Door trim. This dummy hole is resulting into water entry in door panel.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Poor pasting of water shield due to Butyl patch overlap.	Visual check	At Gemba, it was observed that, as Butyl position is shifted downward, watershield gets overlap to Butyl patch	Hypothesis valid

BUTYL-PATCH POSITON FITMENT

While fitment, Butyl positon shifted

Water-shield overlap to Butyl patch as shown

Conclusion:

1.It is observed that,as there is manual fitment of Butyl patch by stage associate. Butyl patch fitment position is shifted.

2.As Butyl-patch position is shifted,while watershield fitment,water-shield adhesive is getting overlap to Butyl patch and watershield is poor pasted.

3.Due to water-shield overlap on Butyl patch,watershield is not getting pasted completely.

4.This result into water leakage. Hence Hypothesis is valid

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Cause	Testing method	Conclusion
1	Water shield flap not inserted	Visual Check	Hypothesis valid
2	Leakage through NVH patch	Visual check	Hypothesis Invalid
3	water shield poor fit	Visual Check	Hypothesis valid
4	Water entry thorugh dummy hole (Model: SCORPIO S3)	Visual check	Hypothesis valid
5	Poor pasting of water shield due to Butyl patch overlap.	Visual Check	Hypothesis valid

5 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTER MEASURE
1	Watershield flap not inserted	Water-shield Flap comes out	Water flap position incorrect	Water flap not locking completely	Correct parts
2	Leakage through NVH patch	Gap observed at topside of NVH patch	NVH patch not getting overlap to inner-panel	NVH rectangular patch width short	Modification of NVH patch
3	water shield poor fit	Stage associate not completely pressed water shield	Due to Inner Door panel deform critical area, water shield is not getting completely pasted	Excess pressure required for water shield butyl sealer sticking	Provide facility for pressing
4	Water entry through dummy hole (Model: SCORPIO S3)	For Scorpio S3-Model plain trim pad-No plug at specific location for lower model	Water entry through dummy hole due to body hole in open condition for trim pad for S3 model	For High Variant there is locator, which is restrict water entry. For low variant_ No locator for door trim	Restrict water entry through dummy hole for low variant door trim
5	Poor pasting of water shield due to Butyl patch overlap.	Butyl position shifted	Manual fitment of Butyl patch	No reference for Butyl patch fitment	Provide reference for Butyl patch fitment

6. Implement the corrective measures

Possible Remedies Identified

Legends

↑ High ↓ Low

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Water flap not locking completely	Correct parts	Low ↓	Low ↓	Yes	Low ↓	Accepted
2	NVH rectangular patch width short	Modification of NVH patch	High ↑	Low ↓	No	Low ↓	Accepted
3	Excess pressure required for water shield butyl sealer sticking	Provide facility for pressing	Low ↓	Low ↓	Yes	Low ↓	Accepted
4	For High Variant there is locator, which is restrict water entry. For low variant_ No locator for door trim	Restrict water entry through dummy hole for low variant door trim	Low ↓	Low ↓	Yes	Low ↓	Accepted
5	No reference for Butyl patch fitment	Provide reference for Butyl patch fitment	Low ↓	Low ↓	Yes	Low ↓	Accepted

6. Implement the corrective measures

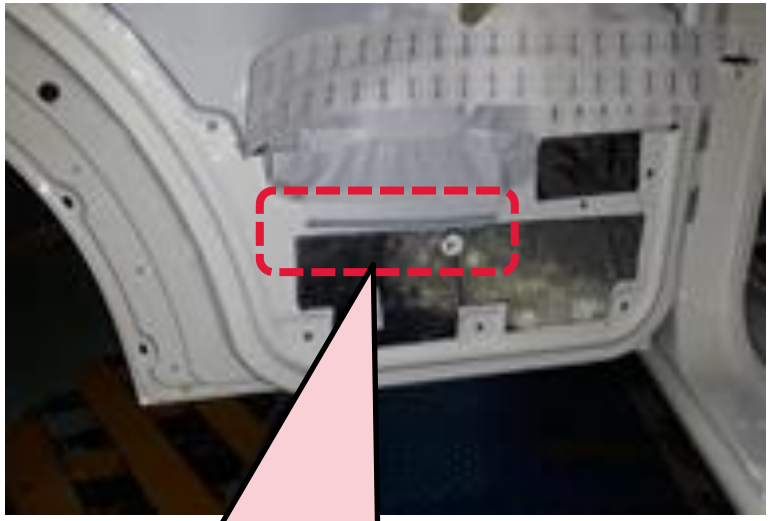
6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Water flap not locking completely	Addition of Butyl sealer on watershield flap,to avoid flap cameout and poor fit	05.09.19	Satish Pagar	Closed

BEFORE

No locking for water-shied flap

AFTER

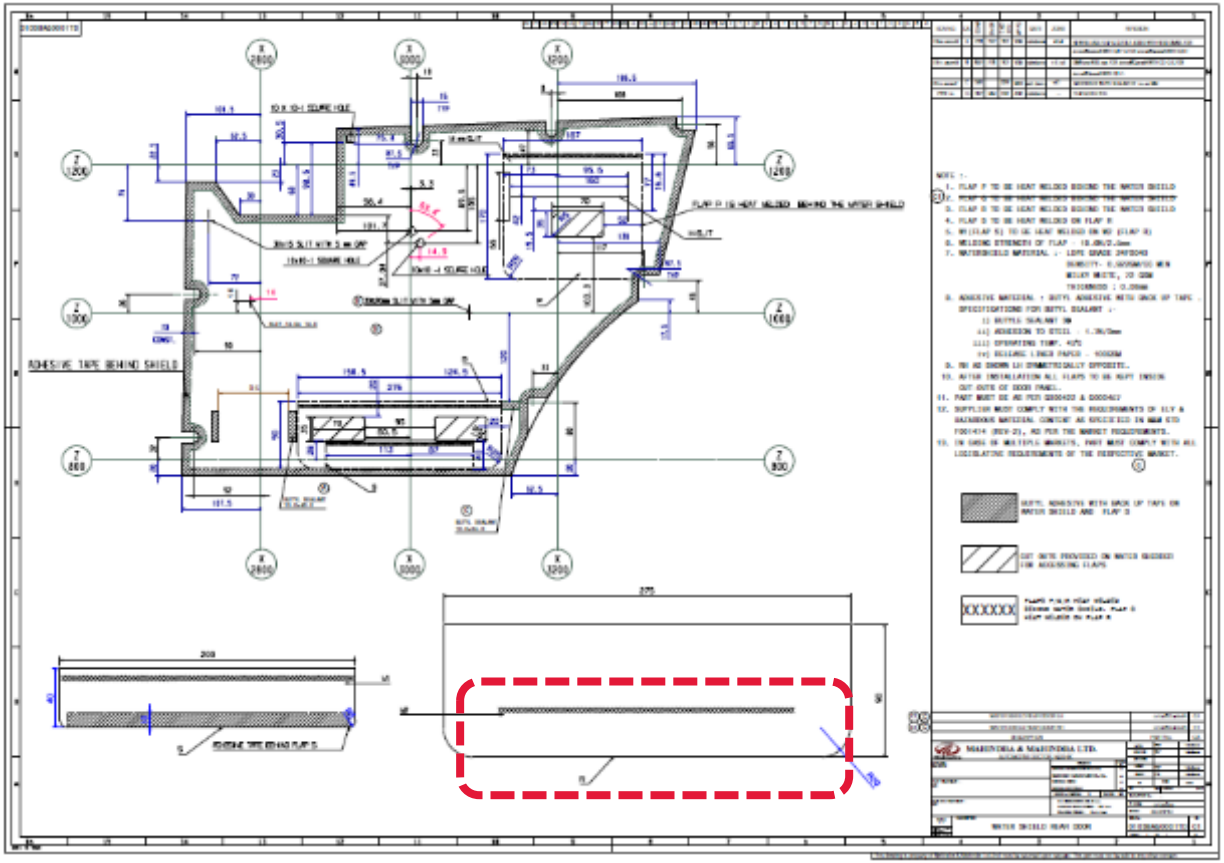


Butyl sealer adhesize added on watershield flap,to avoid flap ameout and poor fit

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
1	Water flap not locking completely	Addition of Butyl sealer on watershield flap,to avoid flap cameout and poor fit	05.09.19	Satish Pagar	Closed



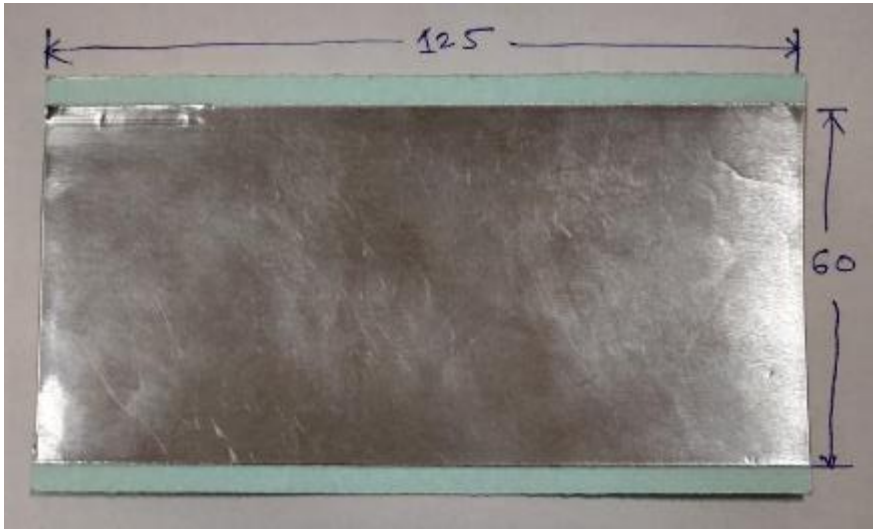
Updated Drawing

6. Implement the corrective measures

6.1 Actions Decided

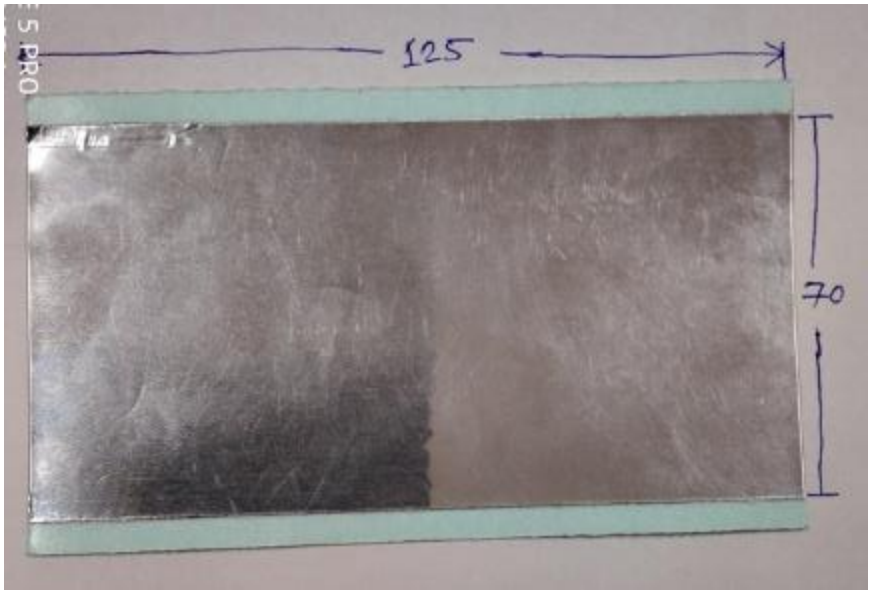
No	Root Cause	Action plan	Actual date	Responsibility	Status
2	NVH rectangular patch width short	Modification of NVH patch-Width of NVH patch increased by 10mm.	10.09.19	Satish Pagar	Closed

BEFORE



Before _ Rectangular Patch 125x60

AFTER

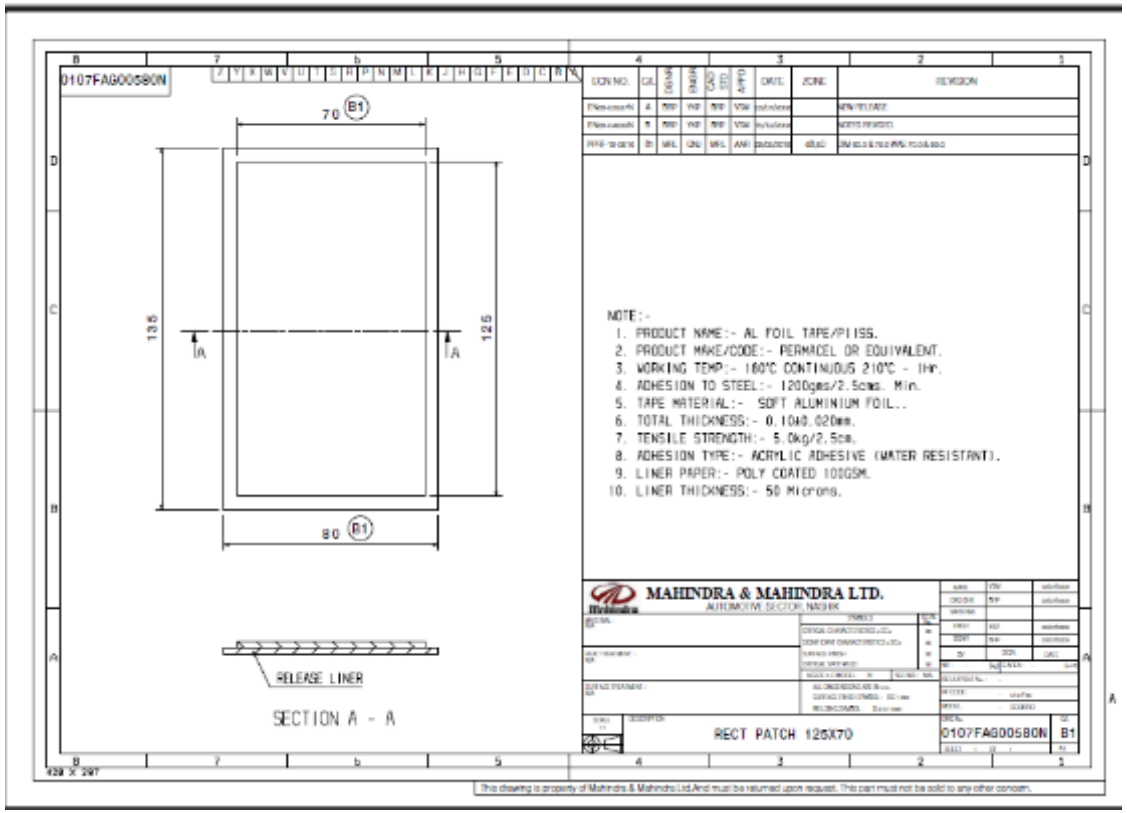


After _ Rectangular Patch 125x70

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	NVH rectangular patch width short	Modification of NVH patch-Width of NVH patch increased by 10mm.	10.09.19	Satish Pagar	Closed



Updated Drawing

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
3	Excess pressure required for watershield butyl sealer sticking	Roller started for water-shield Butyl sealer pressing.	**05.08.19**	Nandy Tarit	CLOSED

BEFORE

Manual pressing of watershield

AFTER

Roller started for water-shield fitment

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
4	For High Variant there is locator, which is restrict water entry. For low variant_ No locator for door trim	Plugging/sticker started to Restrict water entry through dummy hole- for low variants door trim	23.08.19	Nandy Tarit	Closed

Elimination of Water Entry inside Door Panel (Model-Scorpio S3)

BEFORE

NO PLUGGING FOR DUMMY HOLE

AFTER

BLACK STICKER STARTED (AS SHOWN)

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
5	No reference for Butyl patch fitment	Fixture provided for Butyl patch fitment.	23.08.19	Ganesh Bendale	Closed

BUTYL PATCH FIXTURE IMPLEMENTATION

BEFORE

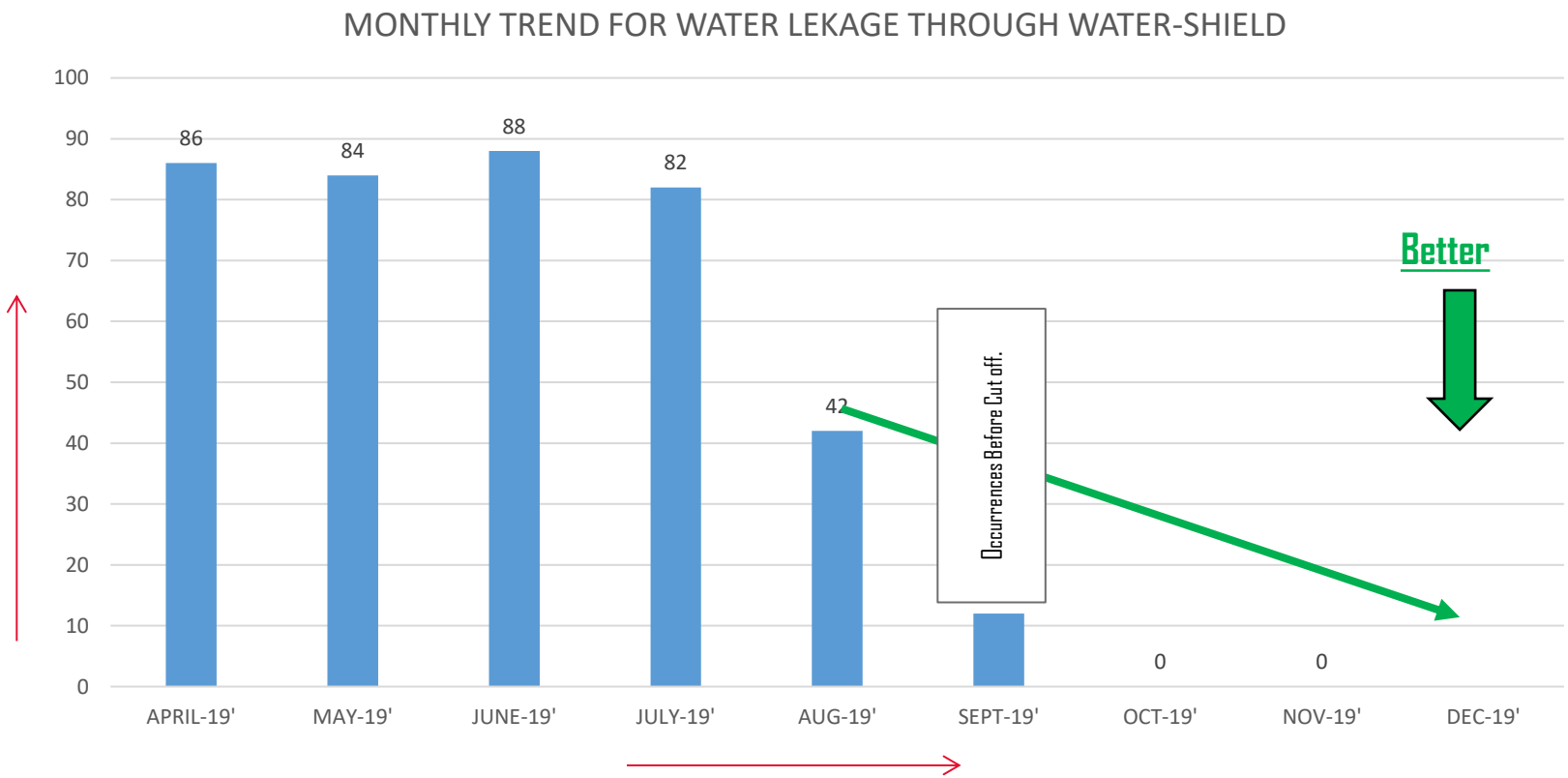
Butyl patch position shifted

AFTER

Reference provided by FRP Fixture implementation9as shown)

7. Check & confirm the results & effects

7.1 Verify Results:



Inference – Concern not reported from after implementing the PCA actions.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S.N	Action Taken	Actual date	Responsibility	Status
1	Watershield Flap modification-Additional adhesize on Flap	05.09.19	Satish Pagar	CLOSED
2	NVH Rectanugular patch correction	10.9.19	Satish Pagar	Closed
3	Roller implementation for watershield fitment	05.08.19	Nanady Tarit	CLOSED
4	Implementation of Butyl Patch fixture	23.08.19	Prasad Deore	Closed
5	PFMEA updation (Rev No –02)	20.09.19	Ganesh Bendale	CLOSED
6	Process audit check sheet updation (Rev No -6)	20.09.19	Sachin Bhamare	CLOSED
7	SOS Updation (Rev No - 02)	25.09.19	Nandy Tarit	CLOSED
8	On job Training given to stage associates for water-shield fitment.	25.06.19	Nandy Tarit	Closed

Check & confirm the results & effects (Cont.....)

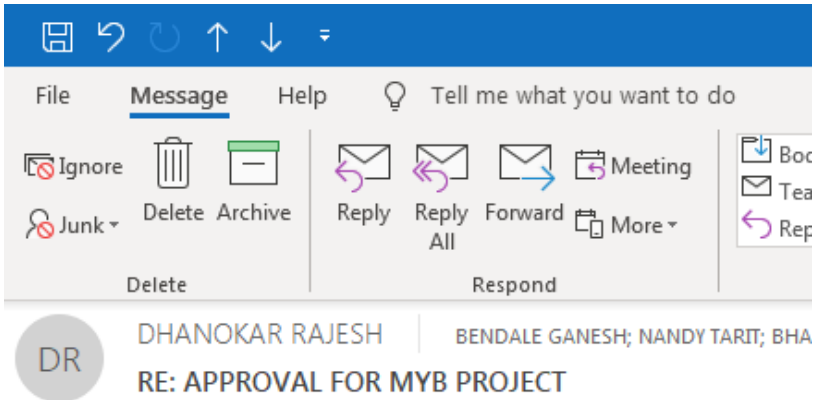
7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of concern water leakage through water shield.
2) 1% Shower FRC improved.
3) Repeat V1 concerns of GNOVAC audit is eliminated completely.
4) Rework cost save upto 61130.3.9 Rs / Year.
5) Shower FRC improved

In tangible Benefits

1) Improved Customer Satisfaction.
2) Operator fatigue reduced , moral improved



Repair /Restoration Cost estimation		
Description	Units	
Rework time (Approx)	Min	12
Avg Concerns/Day	Nos	4.0
Total Required Rework time per day	Min	48
Effective working time	Min	465
Operator required /day	nos	0.10
Approx Operator cost/ day	Rs	1974
Operator cost /concern	Rs	203.77
Avg no. of working days	days	25
Operator cost /month	Rs	5094.2
Operator Cost/year	**Rs**	**61130.3**

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

S.N	Action Taken	Actual date	Responsibility	Status
1	PFMEA updation (Rev No –02)	20.09.19	Ganesh Bendale	CLOSED
2	Process audit check sheet updation (Rev No -6)	20.09.19	Sachin Bhamare	CLOSED
3	SOS Updation (Rev No - 08)	25.09.19	Ganesh Bendale	CLOSED
4	On job Training given to stage associates for Water-shield fitment	25.06.19	Ganesh Bendale	Closed

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job Training given to stage associates	25.06.19	Ganesh Bendale	Closed

8.3 Additional Audit & check :

SN	Document	Target date	Responsibility	Status
1.	Daily Process audit sheet updated	25.09.19	Ganesh Bendale	Closed
2	As per ILU plan, Stage Associate audit done, for adherence to key-point-No NC found	15.10.19	Ganesh Bendale	Closed

8. Standardization

8.1 Updation of documents :-

PFMEA Updation

		Mahindra & Mahindra Ltd.	**Cross Functional Team**
			GANESH BENDALE
Latest Change Letter:	**Optn No.:1800,1810**	**Process Potenial Failure Mode & Effect Analysis.**	Mandar joshi
PU/ Shop :Scorpio TCF	**Optn Desptn: Door Water shield fitment**	**Reviewed by :: GANESH BENDALE** / **Rev. No :** **39** / **Doc No.: PFMEA/SPU/SCORPIO/TRIM/1800_10**	Vikrant Ghule
Module/ Cell :Trim-3	**M/C Name & No.:**	**Approved by :MandarJoshi** / **Rev. Date :** **6.08.19** / **FMEA Date (Orig) :15.11.2005**	Vikrant Morade

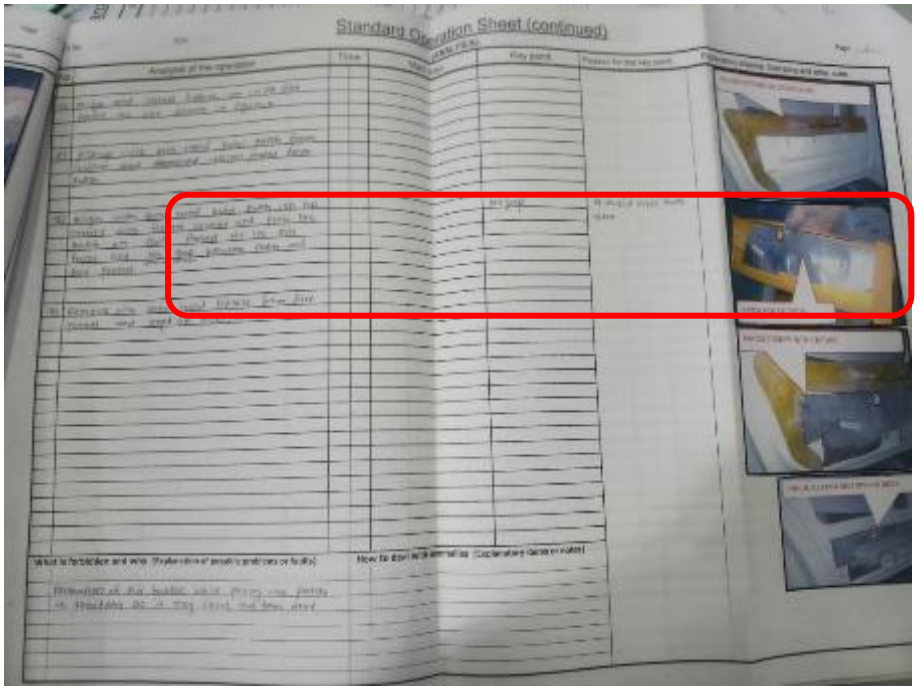
Process step / Function	Requirements	Potential Failure Mode	Potential Effect(s) of Failures	sev	class	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	Occur	Current Process Controls Detection	Detect	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN
Door water shield fitment.	Clipping of speaker socket	Speaker socket clipping not done	**Mfg. & Assy: rework(4) End user: Door noise (4)**	4		Stage associate forgot to clip the speaker scoket	Training to stage associates	3	Self-certification at stage	7	84							
	Mirror socket wire routine	wire not removed through slit provided	**Mfg. & Assy: rework(7) End user:** Water sheild poor pasted (7)	7		Operator not remove the wire from slit provided on w/s	Prtective sleeve provided on ORVM wireloom	2	At stage	7	98							
	Assy of Water shield on Door.	Misalignment of the profile holes on the door/Water sheild come out from door trim **.(Top buyoff concern) Buy Off Concern,17 (GNOVAC concern May 17) (Buy Off concer ,19)**	**Mfg. & Assy:** Water shield will not cover the complete door panel. Rework (7) **End user:** Water may leak, Visible poor aappear.(2) **Mfg. & Assy:** Uneven gap between water shield & door panel. water shield poor fitted. Rework(7) **End user:** Water may leak.(5)	4		Opeartor does not align theshield with body holes properly Operator doesn't follow the proper assy sequence.	Proper fitment procedure is mentioned in SOS.	5	Ensured at same srage.& Water leakage to be check at CAI	7	140							
		Poor fitment / Uneven pressing of shield (RFI concern- Oct08)																
	pressing of water sheild sealer on door panel with Roller ***PCRN/A003/STCF/F18/08764***	ANOVA -C Concern- .JUN'09,Sept09,Oct09 (Top buy off concern in Nov 11) **(NOVAC Jan'13) (CAI top concern,RFI Sept'13,NOVAC Oct13,Buyoff) (RFI Concern,Aug 17) (CAI Concern,19)**	**Mfg. & Assy:** Gap will remain between the shield & door panel. Rework(7) **End user:** Water may leak.(5)	8	**SC**	Opeartor does not press all over the water shield sealer profile (Bottom portion of watershield to be pasted with the help of thumb pressure)	Training to stage associates	6	Water leakage at CAI in shower test.	7	336							
		Water shield flap insertion not done by operator.(RFI Sept'13) **(CAI Concern,19)**	**Mfg. & Assy:** Rework(7) **End user:** Water may leak.(5)	6		Operator does not insert the water shield in door	Trainning to operator 2. Door trim pad fitment operator will ensure the flap insertion & will mark on water sshield before trim pad fitment	6	Check point at CAI shower test	8	288							

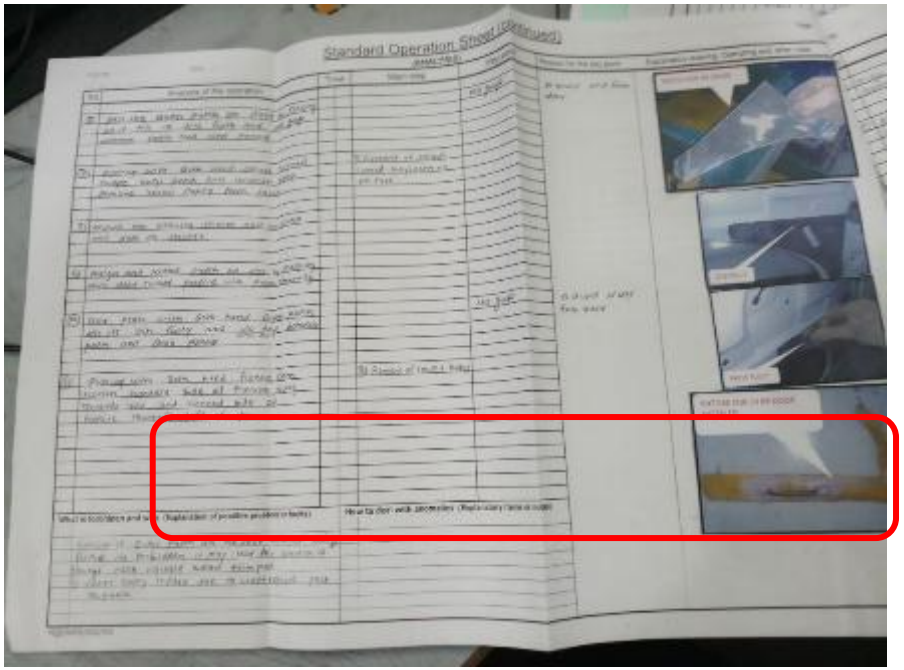
Rev No.	***Rev Date***	***Revision Summary***	***Reason for Revision***
27	30.12.2013	No change	Six monthly review
28	10.10.14	PFMEA Reviewed-For Enchange-32MM Plus shaped slit added on front door watershield & also verftical slit Added	PFMEA updated for EN- Change
29	17.10.14	Occurance rating revised	Six monthly review
30	30.11.15	Occurance rating revised	Six monthly review
31	15.05.16	Occurance rating revised	Six monthly review
32	30.05.2017	GNOVAC CONCERN-Missassembled Water sheild	GNOVAC concern updation may 17
33	30.08.2017	PCRN - Rolle for water sheild pressing	PCRN updation
34	12.10.17	Occurance rating revised	Six monthly review
36	21.02.18	No changes	Six monthly review
37	7.08.18	No changes	Six monthly review
35	1.02.19	GNOVAC CONCERN-Water entry – RH, LH FR & RR door	GNOVAC concern updation
39	6.08.19	No changes	Six monthly review

8. Standardization

8.1 Updation of documents :-

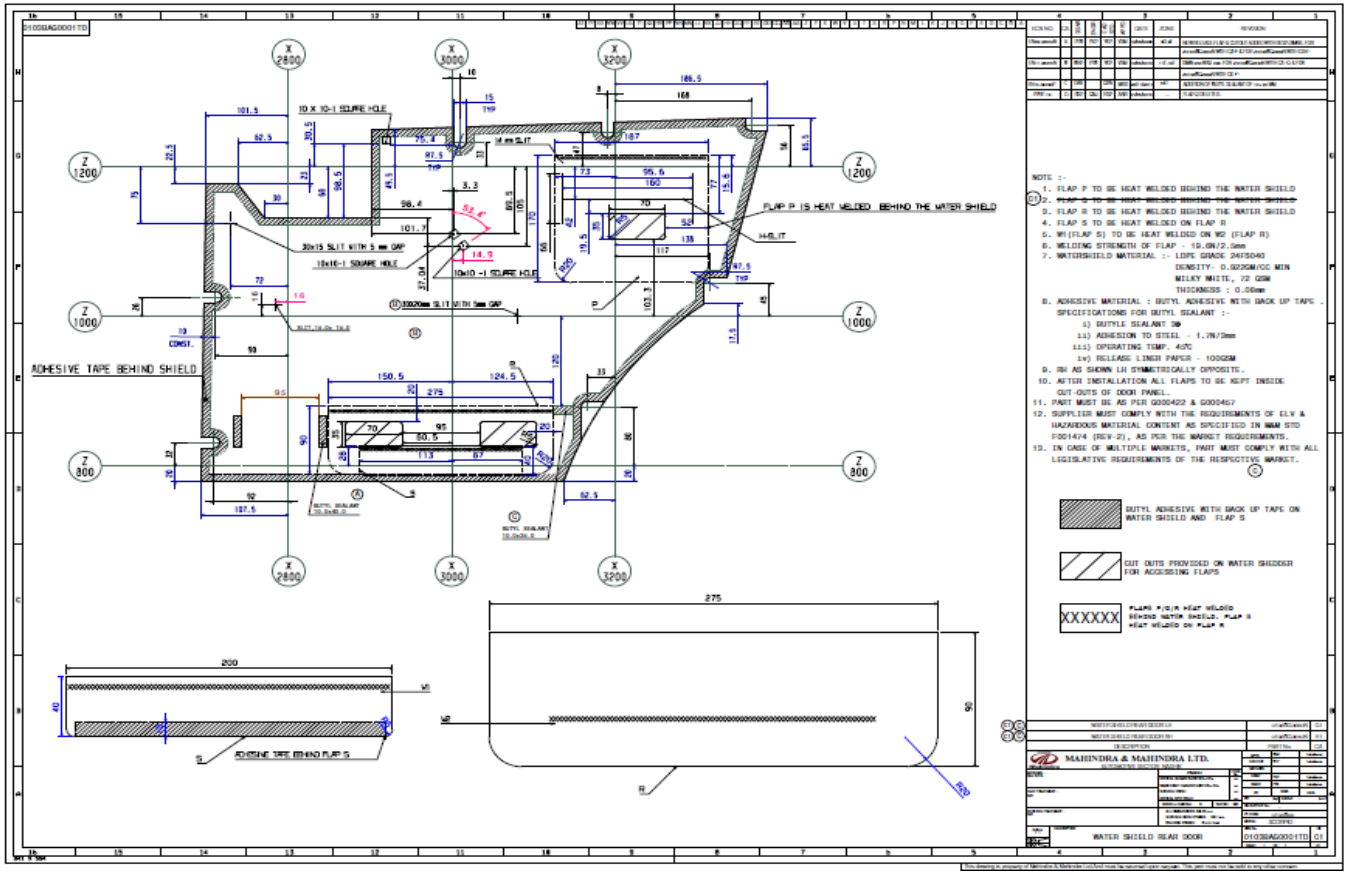
SOS Updation





8.1 Updation of documents : -

DRAWING UPDATION:



8. Standardization

8.1 Updation of documents : -

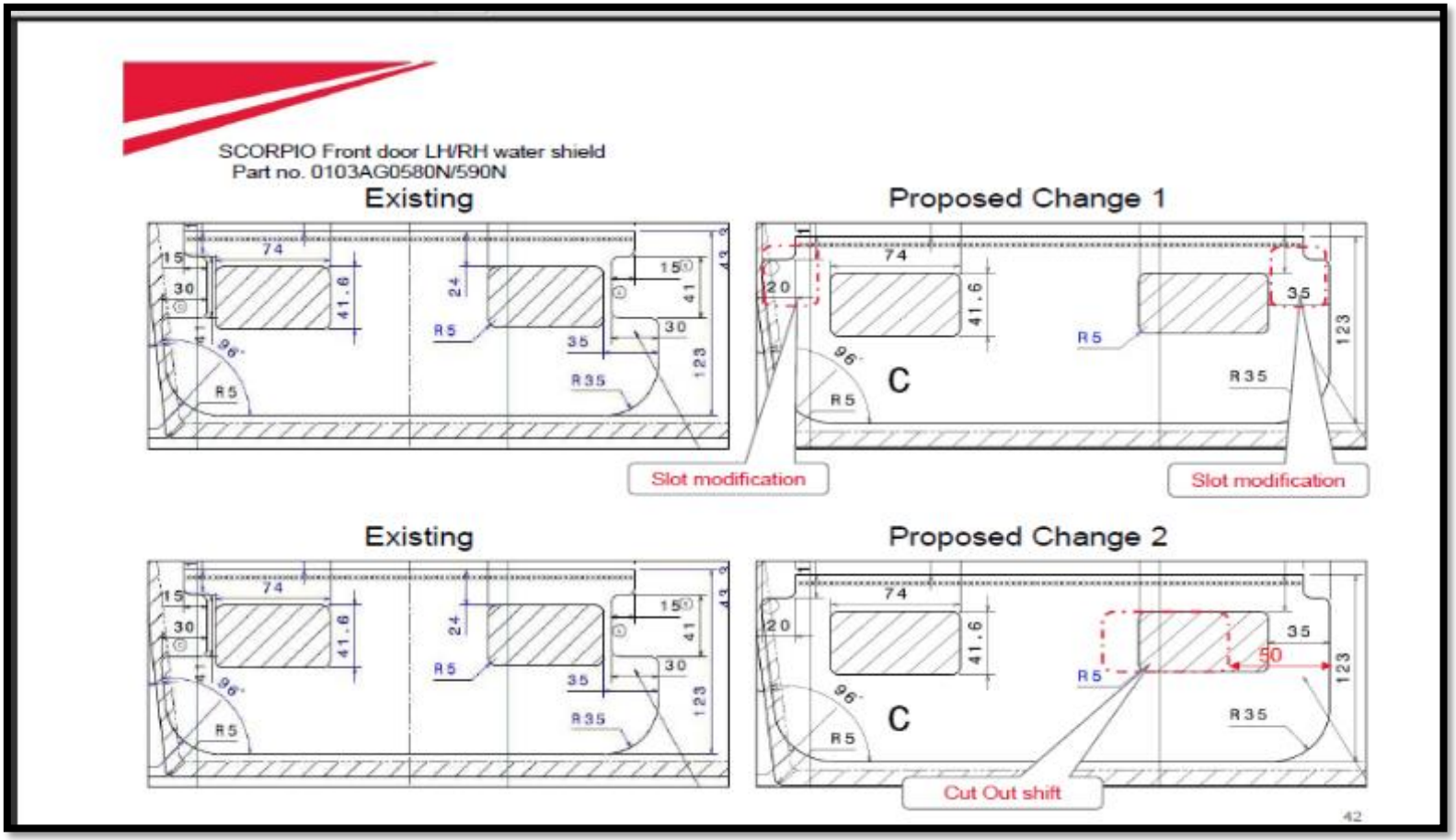
WORK CONTENT: ROLLER STARTED

Sr No	Operation Description	Scorpio Refresh	VLX	Scorpio W110-VLX	Scorpio W110-BSVI	Scorpio W105 - LCCR	Sc W1
8	COVER DOOR RELEASE FITMENT(ALL 4 DOORS) (Only applicable to S3)	0.91	0.91	0.00	0.00	0.91	(
6	RR door Speaker Fitment	0.00	0.00	0.00	0.00	0.00	(
7	WATER SHIELD FRT RH + WATER SHIELD FRT LH WATER SHIELD RR RH + WATER SHIELD RR LH	2.97	2.97	2.97	2.97	2.97	2
	ROLLER APPLICATION ON WATER SHIELD FRONT AND REAR ALL MODEL	0.47	0.47	0.47	0.47	0.47	(
W105	Additional activity of wire routing during front door water shield fitment (LH & RH)	0.00	0.00	0.11	0.11	0.00	(
	DOOR PULL HANDLE FITMENT (RH FRT) + DOOR INNER						

8. Standardization

8.1 Updation of documents : -

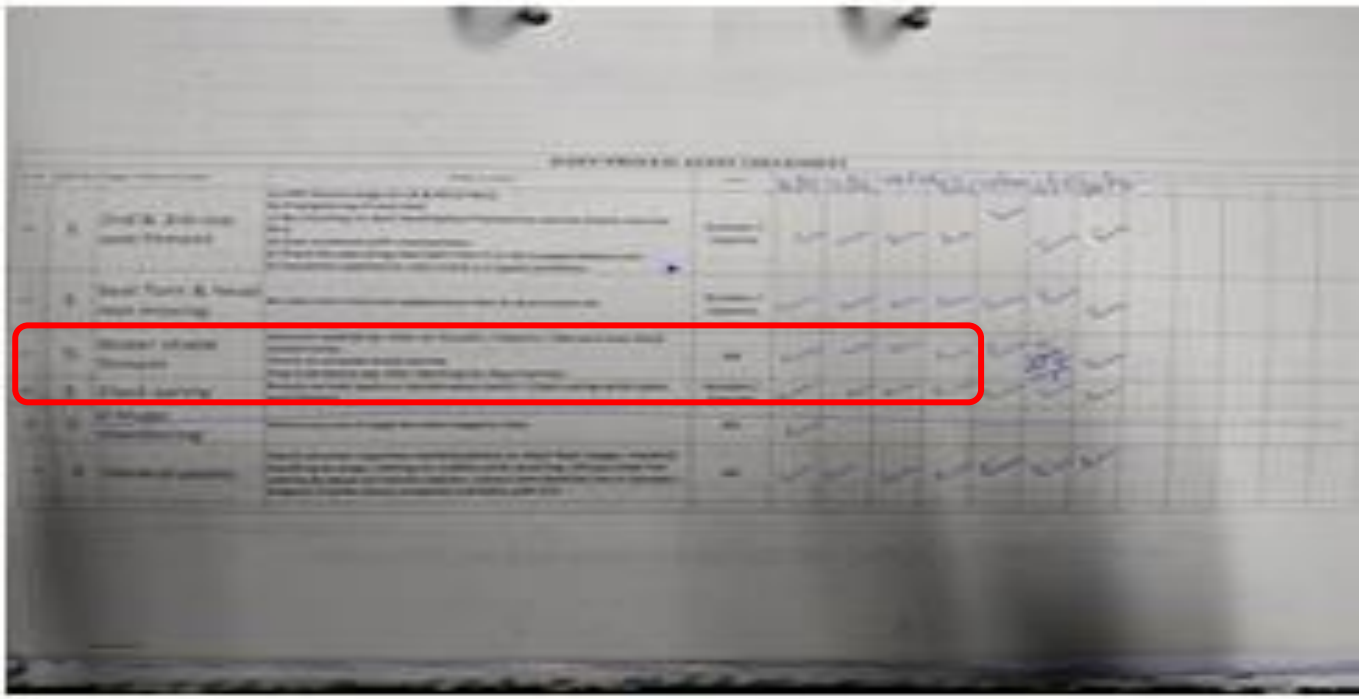
DRAWING UPDATION:



8. Standardization

8.3 Additional Audit & check

Daily Process audit check sheet updation :-



9. Summarize & Plan future actions

9.1 Conclusion:

Water leakage through Concern eliminated after taking actions on watershield modification, Fixture implementation, Providing roller.

9.2 Horizontal Deployment:

For Marazzo veh RH Rear,Lh Frt,Lh Rear Door trim

9.3 Lessons Learnt:

Things Gone Wrong : Failure mode of water-shield flap incorrect position not identified at time implementation.

Things Gone Right : Understood the importance of Watershield pasting on Critical Door panel and its impacts and its mandatory consideration in process.

Thank you

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